■ 2.6 Input and Output

■ 2.6.1 Output Formats

`OutputForm[`*expr*`]`	standard two-dimensional mathematical form
`InputForm[`*expr*`]`	one-dimensional form suitable for input to *Mathematica*

Two basic output formats.

Mathematica usually prints expressions in an approximation to standard mathematical notation. It lays out the parts of the expression in two dimensions, using superscripts, built-up fractions, and so on.

This two-dimensional form is convenient for output, but is not suitable for input to *Mathematica*. The standard input that you type to *Mathematica* consists of lines of text with an effectively one-dimensional structure. Basic *Mathematica* input cannot contain superscripts and so on.

The function `InputForm` allows you to print *Mathematica* expressions in a form that *is* suitable for subsequent input to *Mathematica*.

Here is an expression printed in standard *Mathematica* output form. It involves superscripts and built-up fractions.

```
In[1]:= x^4/4 + 1/y^2

         4
        x     -2
Out[1]= -- + y
        4
```

Here is the same expression in a one-dimensional form suitable for input to *Mathematica*.

```
In[2]:= InputForm[%]

Out[2]//InputForm= x^4/4 + y^(-2)
```

Although the output from a *Mathematica* computation is usually printed in *Mathematica*'s standard output form, many *Mathematica* front ends in fact also store the input form of the *Mathematica* output. (Often they call the input and output forms "unformatted" and "formatted", respectively.) By storing the input forms of all expressions, *Mathematica* front ends allow you to give any expression you see as input to *Mathematica*. A common thing is to look at an expression in output form, then edit the input form of the expression, and use a piece of the text that you get as subsequent input to *Mathematica*.

Section 1.9 discussed some of the ways you can generate output from *Mathematica* that is suitable for input to other computer languages and programs.

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`TeXForm[`*expr*`]`	TeX input form
`CForm[`*expr*`]`	C language input form
`FortranForm[`*expr*`]`	FORTRAN input form

Mathematica output suitable for input to other systems.

Here is an expression in standard *Mathematica* output form.

```
In[3]:= x^5/7 + y^3 - Log[y]

          5
         x     3
Out[3]= -- + y  - Log[y]
         7
```

Here is the same expression in a form suitable for input to the TeX typesetting system.

```
In[4]:= TeXForm[%]

Out[4]//TeXForm= {{{x^5}}\over 7} + {y^3} - \log (y)
```

Here is the expression in C language form. C macros for objects like `Power` are defined in the header file `mdefs.h`.

```
In[5]:= CForm[%]

Out[5]//CForm= Power(x,5)/7 + Power(y,3) - Log(y)
```

Here is the FORTRAN form.

```
In[6]:= FortranForm[%]

Out[6]//FortranForm= x**5/7 + y**3 - Log(y)
```

Section 1.9 discussed some of the details of TeX, C and FORTRAN output in *Mathematica*, and outlined how you can take this output, and "splice it" into external files.

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